

CSE 4891 – Data Mining

Part A: Introduction

1	Course Title	Data Mining
2	Course Code	CSE 4891
3	Pre-requisites	CSE 3841
4	Course Type	Elective Course
5	Credit Hours	3.00
6	Contact Hours	3 Hours/Week
7	Semester	8 th
8	Total Marks	100
9	Course Instructor's Information	Md. Mushfiqul Haque Email: mushfiqul@cse.uiu.ac.bd Office: 837 (A)
10	Course Rationale	Data Mining is currently regarded as the key element of a more general process called Knowledge Discovery that deals with extracting useful knowledge from raw data. The knowledge discovery process includes data selection, cleaning, coding, using different statistical and machine learning techniques, and visualization of the generated structures.
11	Course Objectives	The objectives of this course are: • To introduce students to the basic concepts and techniques of Data Mining. • To develop skills of using recent data mining software for solving practical problems. • To gain experience of doing independent study and projects.

Part B: Content of the Course

12	Course Contents (approved by UGC)	Introduction to data mining: data mining task and applications, data preprocessing, feature selection, association analysis, frequent item-set mining; Single model classifier: NN, k-nearest neighbor, naive Bayes classifier, decision tree induction, naive Bayesian tree, rule-based classifier; Model evaluation and selection; Ensemble learning: random Forests, bagging, boosting, isolated forests; Clustering: k-means clustering, similarity-based clustering, nearest-neighbor clustering, density-based clustering, ensemble clustering, evaluation of clustering methods, clustering high-dimensional data; EM algorithm, PCA algorithm, Data balancing methods; Active learning; Neural network, Perceptron.	
13	Course Outcomes (COs)		
		COs	Description

		CO1	Understand a wide range of emerging methodologies and technologies to
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facilitate knowledge discovery.

CO2

Understand discovering and measuring interesting patterns from different kinds of datasets.

CO3 CO4

Understand data classification methods.

Learn standard data clustering techniques.

14 Mapping of COs and Program outcomes

COs	Program Outcomes(POs)									
	PO1	PO2	PO3	PO4	PO5 PO6 PO7 PO8	PO9	PO10	PO11	PO12	
C	C									
O				C						
1		C C								
C										
O										
2										
C										
O										
3										
C										
O										
4										

15 Mapping COs with Teaching-Learning and Assessment Strategy

Class/Week	Topics/Assignments	Course Reading Outcomes (COs)	Reference	Teaching- Learning Strategies	Assessment Strategies
1	Introduction to Data Mining Related technologies, Origin of Data Mining, Goals, Data mining Tasks (predictive and descriptive)	CO1	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam

			Book		
5	Regression(linear, multi-variable , polynomial, logistics)	CO1		Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
6	Regression(linear, multi-variable , polynomial, logistics), (Cond.)	CO1		Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
6	Entropy, Gain, Decision tree, ID3 algorithm, Confusion matrix, Precision, Recall, F1-factor, Accuracy, error rate, ROC curve.	CO1	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
7	Review and Class Test/Quiz	CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignment s/Quizzes/Exam
8	Simple decision Tree, Algorithm: ID3, CART algorithm, C 4.5 algorithm	CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
9	Simple decision Tree, Algorithm: ID3, CART algorithm, C 4.5 algorithm (cond.)	CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
10	Confusion matrix and Recall, Precision, True positive rate, False positive rate, Entropy and Gain	CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
11	Basic issues Training and testing(open test close test) Estimating classifier accuracy (holdout, cross-validation-fold,Stratified, LOOC)	CO1, CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
12	Data mining memory based algorithms: NN, KNN, Mahalanobis distance	CO3		Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
13	Association rules, Rule support, Rule confident, Rule coverage, Rule Induction, Rule mining and Deduction	CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
14	Understanding Apriori algorithm and demonstrate association mining	CO2	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
15	Ensemble learning concept, Combining multiple models (bagging, boosting, stacking)	CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam

16	Clustering Algorithms: k-means, Modified K-means	CO3		Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
17	Hierarchical clustering, Dendrogram	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
18	DBScan clustering, Conceptual clustering: Cobweb	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
19	Kmedoids, Expectation maximization (EM)	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
20	Principal component analysis (PCA), Page rank algorithm	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
21	Bayes theorem (math), Naïve bias theorem (math) [Classification Technique]	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
22	Neural network, Perceptron Activation functions (Threshold Piecewise, Sigmoid, Signum function), Stochastic model of a neuron	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
23	NN with single layer and two layers, NN without bias and with biases, Back-propagation	CO4	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
24	Presentaton/Experiments with SKlearn toolkit – Clustering and classification algorithms	CO4	Lecture Slides and Text/ Ref. Book/ Online Materials	Lecture/Group Discussion/Exercise	Class Tests/Assignments/Quizzes/Exam
25	FINAL EXAM				

Part C: Assessment and Evaluation Methods

Assessment Strategy	Assessment Types	Marks
Formative Assessment	Attendance	5%
	Assignments	5%
	Term Paper, CT	20%
Summative Assessment	Mid Term	30%

	Final Exam	40%
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Part D: Learning Resources

Text Book	Data Mining: Concepts and Techniques by Jiawei Han, Micheline Kamber and Jian Pei, 3 rd Edition, Elsevier, 2011.
Reference	Pang-Ning Tan, Michael Steinbach, Vipin Kumar - Introduction to Data Mining- Pearson (2013) Data Mining and Machine Learning: Fundamental Concepts and Algorithms Second Edition, Mohammed J. Zaki and Wagner Meira, Jr, Cambridge University Press, March 2020
LMS URL	